NATURAL DISASTERS IN INDIA

Hazards | Vulnerability | Major Events | Response

59%
of India's land is prone to earthquakes of moderate to very high intensity

12%
of land (over 40 million hectares) is prone to floods and river erosion

5,700 km
of the 7,516 km coastline is prone to cyclones and tsunamis

68%
of the cultivable area is vulnerable to drought

Source: National Disaster Management Authority / Ministry of Home Affairs

1. TYPES OF NATURAL DISASTERS

GEOLOGICAL

* Earthquakes
* Landslides
* Tsunami
* Mining-related hazards

Example: Bhuj (Gujarat) earthquake, 2001

HYDRO-METEOROLOGICAL

* Floods and flash floods
* Cyclones and storm surges
* Droughts
* Lightning, thunderstorms, hailstorms
* Avalanches
* Cold waves and heat waves

Example: Cyclone Fani (Odisha), 2019

BIOLOGICAL

* Epidemics
* Pest attacks (e.g. locust swarms)
* Forest fires

Example: Desert locust invasion, 2019-20

2. MAJOR NATURAL DISASTERS IN INDIA

Year	Event	Impact
1905	Kangra earthquake (Himachal Pradesh)	About 20,000 deaths
1993	Latur earthquake (Maharashtra)	Nearly 10,000 deaths
1999	Odisha Super Cyclone	About 10,000 deaths
2001	Bhuj earthquake (Gujarat)	About 20,000 deaths; led to creation of GSDMA
2002	Nationwide drought	Around 300 million people affected
2004	Indian Ocean tsunami (26 December)	Struck Tamil Nadu, Andhra Pradesh, Kerala and the Andaman & Nicobar Islands
2013	Uttarakhand flash floods	Thousands dead or missing; Kedarnath region devastated
2018	Kerala floods	Most severe floods in the state in about a century
2019	Cyclone Fani (Odisha)	About 1.2 million people evacuated, limiting casualties

3. WHY IS INDIA SO VULNERABLE?

* The Indian plate presses north into the Eurasian plate, making the young Himalayas quake- and landslide-prone.
* A 7,516 km coastline with two cyclone seasons exposes the coast to storm surges and tsunamis.
* Dependence on the monsoon: floods in one region and drought in another in the same year.
* High population density, rapid urbanisation and weakly built houses increase losses.
* Deforestation, encroachment of river beds and climate change make extremes more frequent.

4. EARTHQUAKE SEISMIC ZONES

ZONE V	Very high risk
ZONE IV	High risk
ZONE III	Moderate risk
ZONE II	Low risk

Zone V includes Kashmir, parts of Himachal Pradesh and Uttarakhand, the North-East, the Rann of Kutch and the Andaman & Nicobar Islands.

5. HOW INDIA RESPONDS

Disaster Management Act, 2005
Legal framework for prevention, mitigation, preparedness and response

NDMA / SDMA / DDMA
National (PM), State (CM) and District authorities plan and coordinate

Early warning
IMD issues cyclone and weather warnings; CWC issues flood forecasts

NDRF
Specialised National Disaster Response Force for search, rescue and relief

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Sources: NDMA; Ministry of Home Affairs; Lok Sabha reply on disasters (3 Dec 2014); ADRC India country reports.